

An Obstruction Calculus for Viability under Incomplete Observation

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September 10, 2026

Abstract

Under perfect measurement the viability kernel — the set of states from which some feedback keeps the system within its constraints — is characterized by tangency conditions. Under incomplete observation the sufficiency direction has a canonical answer in Veliov’s output-feedback regulation condition and the estimation-tube reduction. The complementary direction — certifying that no observation-based policy is viable — has received less systematic treatment.

This paper develops that instrument: a calculus of obstruction certificates, sound sufficient conditions for nonviability (equivalently, necessary conditions for viability), finitely checkable in the common-action, fibre-certification, and finite-horizon forms and analytic drift-and-timing conditions elsewhere. They do not exhaust the complement of the epistemic kernel, the kernel’s observation-based counterpart; in finite systems backward recursion is complete. Five mechanisms are established — two finitely checkable, two closed-form conditional, one minimal — with a sixth under a policy-class restriction. The central common-action obstruction shows that when the safe controls of compatible states intersect emptily, no observation-based policy is viable, though every compatible state is individually viable under full information. The others are a finite-time exit certificate under an Isaacs-type drift condition, an epistemic-emptiness construction in which a constant observation merges states with incompatible admissible controls, a delayed-information obstruction with an explicit timing bound, and a certification limit: an exact observation-only certifier exists iff safe-set membership is constant on the observation fibres. The sixth, a certainty-equivalence trap, empties the kernel of the uncorrected controller under a biased observation even when the perfect-information kernel is nonempty.

The calculus is positioned against barrier certificates and estimation tubes, and its monitoring and institutional consequences — observation timing, coarseness, aggregation, and bias — are drawn.

Keywords: viability theory; viability kernel; obstruction certificates; robust viability; epistemic viability; output feedback; sustainability governance; information structures; partial observation

Mathematics Subject Classification: 49J53; 93B03; 93C41; 91B76

1. Introduction

1.1 The problem

Viability theory characterizes the states of a constrained control system from which at least one control can keep every future state within a constraint set (Aubin, 1991). For sustainability problems the constraints are floors — stock levels, service thresholds, safety margins — and viability is the formal counterpart of maintaining a development path rather than merely optimizing it (Béné, Doyen, and Gabay, 2001; De Lara and Doyen, 2008; Doyen et al., 2012; Doyen and Gajardo, 2020). The theory under *perfect measurement* is complete in the relevant sense: the viability kernel exists as the largest viable subset, and Nagumo-type tangency conditions certify it.

Sustainability governance, however, operates under *incomplete observation*. Stocks are assessed at discrete review intervals. Indicators are coarser than the state. Some stocks are unobserved altogether. And the information that does arrive may arrive late. This paper addresses that side: for an incomplete observation structure, it develops instruments that certify that no observation-based policy is viable — that the infeasibility is due to the information structure, not to the dynamics.

Floors in sustainability assessment are typically constraints on the productive base — the state of the system that yields the measured output — rather than on the output itself. Such a base can be read as natural capital, as a stock, or as a slowly regenerating flow of services; these are overlapping readings of the same asset rather than mutually exclusive ones, and what makes a use of it sustainable is whether the use falls on the yield or on the base itself. Across assets, whether the base behaves as a flow or as a stock is a continuum, set by its regeneration timescale relative to the rate of use — a continuum that runs from a season, through a year, to geological time for a mineral deposit. If a base regenerates too slowly for the rate at which it is taken, drawdown is still liquidation.

A base can also erode while measured output is maintained: the degradation is not yet reflected in the measured quantity, or it is offset by a higher per-unit service, because use may draw on the base rather than on its yield. The signal that would reveal the erosion is accordingly often the one that is coarse, postponed, or absent. Incomplete observation is therefore structural to the certification problem, not merely a practical limitation; the sufficiency literature and the obstruction calculus of this paper address the same underlying indeterminacy rather

than competing ones.

The question splits naturally. The sufficiency direction has a canonical literature: Veliov (1993) gives conditions for the existence of an output-feedback regulation map under incomplete and inexact measurement, reducing under perfect measurement to the classical viability condition (Haddad, 1981); the estimation-tube programme (Cardaliaguet, Quincampoix, and Saint-Pierre, 2007) absorbs imperfect information into an estimation space of sets of compatible states with no loss of value; and barrier certificates certify safety (Prajna and Jadbabaie, 2004; Prajna, Jadbabaie, and Pappas, 2007), with converse and hybrid necessary-and-sufficient characterizations (Prajna and Rantzer, 2005; Maghenem and Sanfelice, 2019). What that literature does not supply is the necessity side: a calculus of obstruction certificates. An obstruction certificate is a checkable witness that no observation-based policy exists — finite in the finite-state and polyhedral cases — and the argument it makes is that a prescribed class of observation-based policies fails not because a particular policy is bad, but because the information structure leaves no room for any policy. In the polyhedral common-action and certification forms (Theorem 2 and Proposition 6) it is a finite, checkable test; the drift certificates of Theorem 4 and Theorem 3 are not finite objects, one exhibiting an enforcement selection and the other bounding the timing uniformly over every policy. Veliov’s condition tells us when output feedback can work; the obstruction calculus tells us when it cannot, and for the borderline cases it identifies which quantitative feature of the observation design — timing, coarseness, bias, aggregation — is responsible.

1.2 Contributions

Five obstruction mechanisms are developed, each with a complete proof, and a sixth is exhibited under a policy-class restriction:

1. **The instantaneous common-action obstruction** (*finitely checkable*) (Theorem 2, Example 1). If the safe-control sets of the states compatible with the current information set intersect emptily, then no observation-based policy is viable, even though every compatible state is individually viable under full information. The failure is purely informational: no stochasticity, no estimation error.
2. **The delayed-information obstruction** (*closed-form conditional*) (Theorem 3). If the information needed to distinguish safe from unsafe compatible states arrives only at time T_{obs} , and the disturbance can force constraint violation earlier than T_{obs} , then no observation-based policy is viable. The timing bound $\inf q/\varepsilon < T_{\text{obs}}$ is stated explicitly: information may be accurate but arrive too late.
3. **The finite-time exit certificate** (*closed-form conditional*) (Theorem 4). Under a Dini-drift condition of Isaacs type — the disturbance chooses after the control — the disturbance can force exit from the safe set within an explicit time bound, for *every* admissible control of any information structure. This is the base obstruction: when it applies, the viability question is closed without any observation-theoretic argument.

4. **Epistemic emptiness by admissibility** (*minimal construction*) (Proposition 5, a minimal construction). A constant observation merges states whose *admissible* controls differ: the state-dependent control set forces a single common action, and that action is unsafe at the boundary, so the unresolved initial observation fibre lies outside the epistemic kernel, even though every physical state is viable under full information. The mechanism is admissibility — the common-action obstruction of Theorem 2 at minimal size — and Example 1 gives the hidden-mode instance.
5. **The fibre certification criterion** (*finitely checkable*) (Proposition 6, Corollary 1). An exact observation-only safety certifier — a function of the observation that returns “safe” exactly on the safe set — exists if and only if safe-set membership is constant on every observation fibre. The certainly-safe set is the sound relaxation: the largest set of observations that can be labeled safe without further information.
6. **The certainty-equivalence trap** (*policy-class restriction*) (Remark 3). Even an *injective* observation empties the kernel of a system whose perfect-information kernel is nonempty for the canonical certainty-equivalence controller — the perfect-information zeroing law applied to the uncorrected observation. The trap is a policy-specific failure: a controller that uses the observation map’s structure (the bias correction) restores viability.

The calculus is organized by a single selector principle (Proposition 1, Section 2.4): observation-based viability is the existence of one response — admissible, safe, and recursively viable — for every compatible state, and each certificate proves that a specific response correspondence is empty; the obstructions are nested in a ladder (Proposition 3). Five refinements complete the picture: backward belief recursion is sound and complete on any finite horizon in finite systems (Theorem 1); a uniform-margin condition connects the instantaneous and tube obstructions (Proposition 2); the timing obstruction has a sharp threshold form (Remark 1); a Helly-type sparse witness bounds the common-action obstruction by $m + 1$ compatible states (Proposition 4); and a comparison-function form sharpens the exit-time bound (Remark 2). The epistemic kernel is monotone in action sets, disturbances, information, and policy class (Proposition 7).

Throughout, “obstruction” means a *certified* failure. Theorem 4 and Theorem 3 each exhibit the violating constraint, an admissible disturbance, and the quantitative bound that make failure inevitable, for every policy in the prescribed class; Proposition 5 and Theorem 2 exhibit the incompatible admissible controls — the merged observation whose incompatible safe controls certify emptiness — with no quantitative bound claimed there; and Proposition 6 is the converse-side characterization of the certification limit itself. The calculus is positioned against the barrier-certificate literature (Section 6.2) and the estimation-tube programme (Section 6.3), and its consequences for the design of monitoring and indicators in sustainability governance are drawn in Section 6.4.

1.3 Related work

The viability-theory background is Aubin (1991), Aubin, Bayen, and Saint-Pierre (2011), and the kernel approximation theory of Saint-Pierre (1994); robust viability is Aubin and Frankowska (1990) and Frankowska (1989). Under incomplete measurement, the sufficiency theorem is Veliov (1993), viability with a priori unknown but observable parameters is Quincampoix and Veliov (1994), and the estimation-set reduction is Cardaliaguet, Quincampoix, and Saint-Pierre (2007). On the failure side, Aubin (2001) characterizes the complement of the kernel — Poincaré’s “shadow” — together with the capture basins, and Aubin and Catté (2002) supply the bilateral-fixed-point and discriminating-kernel calculus; the certificates of Section 3 are one-sided, observation-constrained analogues of that programme. In verification, barrier certificates originate with Prajna and Jadbabaie (2004), with worst-case and converse results in Prajna, Jadbabaie, and Pappas (2007) and Prajna and Rantzer (2005) and hybrid characterizations in Maghenem and Sanfelice (2019). The sustainability application domain is Béné, Doyen, and Gabay (2001), De Lara and Doyen (2008), Doyen et al. (2012), and Doyen and Gajardo (2020). To our knowledge, the obstruction certificates of Theorem 2, Theorem 3, and Proposition 6 — in particular the certification criterion and the timing bound — have not been stated in this form; the elementary facts on which they rest (quantifier commutation and Dini comparison) are classical.

1.4 Organization

Section 2 fixes the framework: the control system, the viability kernel hierarchy, observation structures, information states, the safe-control correspondence, and the selector principle (Proposition 1). Section 3 develops the obstruction calculus (Theorem 1–4, Propositions 2–5, Remarks 1–2, and Example 1): finite-horizon completeness, the instantaneous common-action obstruction with its ladder and sparse witness, the delayed-information obstruction with its threshold form, and — as supporting certificates — the finite-time exit certificate and the admissibility obstruction. Section 4 develops the certification limits (Proposition 6, Corollary 1, Remark 3). Section 5 reviews the sufficiency landscape, cited rather than re-derived. Section 6 discusses the position of the calculus relative to barrier certificates and estimation tubes, states the monotonicity of the epistemic kernel (Proposition 7), and draws the governance consequences. Sections 7–9 develop the new material: complete characterizations and the residual gap (Section 7), a calibrated case study with a coverage audit (Section 8), and the probabilistic belief-state development (Section 9). Section 10 concludes. Proof sketches of the Section 3 results are given in the main text; the complete proofs of the Section 3 certificates, the expanded sufficiency review, and the auxiliary constructions are collected in the Supplementary Material; the results of Sections 7 and 9 are proved in the main text.

2. Framework

2.1 The control system and the kernel hierarchy

This section fixes the mathematical objects used throughout the paper: the control system, the disturbance structure, the constraint set, and the four kernels (perfect-information, robust, epistemic, and institutionally restricted) that organize the analysis.

Fix a state space $X \subseteq \mathbb{R}^n$, a control set U , a disturbance set D , and dynamics

$$\dot{x}(t) = f(x(t), u(t), d(t)), \quad u(t) \in U(x(t)), \quad d(t) \in D(x(t)),$$

with f continuous, U and D set-valued with closed graph and nonempty compact values, and admissible controls and disturbances taken to be measurable selections — the standing solution-existence conditions: every selection pair admits a Carathéodory solution, and the convexified relaxed inclusion of Section 3.5 satisfies the Marchaud conditions of the viability literature (Aubin, 1991). Let $\mathcal{V} \subseteq X$ be a closed constraint set. A *causal policy* π is a map from the observation record to controls, respecting the specified information structure. The standard objects are:

- $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}})$: the **viability kernel** under full-information (state-feedback) policies — the largest closed subset of $\mathcal{V}$ from which there exists a state-feedback control keeping the trajectory in $\mathcal{V}$ for all time — no disturbance, or a known disturbance trajectory (Aubin, 1991).
- $\text{RViab}(\mathcal{V})$: the **robust kernel**, the largest subset of $\mathcal{V}$ from which there exists a state-feedback policy keeping every trajectory — for *every* admissible disturbance realization — in $\mathcal{V}$ (Aubin and Frankowska, 1990).
- $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$: the **robust epistemic kernel** — the object of this paper (Definition 1). Its elements are *states of information*, not physical states (Section 2.3). The non-robust counterpart $\text{EViab}_{\mathcal{I}}(\mathcal{V})$ is the non-robust contrast class (Section 2.3).

Projected to physical state space — with $\text{proj}_{\mathfrak{B}}(\mathfrak{B}) = \bigcup_{B \in \mathfrak{B}} B$ the *existential* projection of a collection $\mathfrak{B}$ of information states, applied to the first two terms, whose elements are information states — the informational hierarchy reads

$$\begin{aligned} \text{proj}_{\mathfrak{B}}(\text{IRViab}_{\mathcal{I}}(\mathcal{V})) &\subseteq \text{proj}_{\mathfrak{B}}(\text{ERViab}_{\mathcal{I}}(\mathcal{V})) \\ &\subseteq \text{RViab}(\mathcal{V}) \subseteq \text{Viab}(\mathcal{V}), \end{aligned}$$

where $\text{IRViab}_{\mathcal{I}}(\mathcal{V})$ is the institutionally restricted kernel of Section 6.4. The physical implication of epistemic viability is the inclusion at the level of beliefs: $B_0 \in \text{ERViab}_{\mathcal{I}}(\mathcal{V}) \Rightarrow B_0 \subseteq \text{RViab}(\mathcal{V})$. Each inclusion can be strict, with a distinct cause: robust contraction arises from disturbances; epistemic contraction from indistinguishability; institutional contraction from restricted authority, enforcement, and allocation. The present paper characterizes the *epistemic* contraction: the mechanisms by which indistinguishability empties or shrinks the kernel.

2.2 The safe-control correspondence

For a constraint set $\mathcal{V}$ described by finitely many C^1 constraint functions, $\mathcal{V} = \{x : q_j(x) \geq 0, j = 1..m\}$, define the **safe-control correspondence** at $x \in \mathcal{V}$:

$$\begin{aligned} \mathcal{R}_{\mathcal{V}}(x) = \{ &u \in U(x) : \forall d \in D(x), \forall j \text{ with } q_j(x) = 0, \\ &\nabla q_j(x) \cdot f(x, u, d) \geq 0 \}, \end{aligned}$$

the set of controls whose drift is inward (subtangential) on every active constraint face, uniformly over the disturbance. For x in the interior of $\mathcal{V}$ (no active constraints), $\mathcal{R}_{\mathcal{V}}(x) = U(x)$. Two levels of the tangency condition are distinguished throughout. (i) *Local reading*. The correspondence is applied to the

constraint set itself: by Nagumo’s theorem (Nagumo, 1942) in its robust form, if $\mathcal{V}$ is closed and $\mathcal{R}_{\mathcal{V}}(x) \neq \emptyset$ on $\partial\mathcal{V}$ with a measurable selection $u(x) \in \mathcal{R}_{\mathcal{V}}(x)$, then $\mathcal{V}$ is itself robustly invariant, i.e. $\text{RVIab}(\mathcal{V}) = \mathcal{V}$ (Aubin, 1991; Frankowska, 1989). (ii) *Kernel reading*. When $\mathcal{V}$ is not invariant, viability of a point $x_0 \in \mathcal{V}$ requires instead a selection $u(x) \in \mathcal{R}_G(x)$ along the trajectory, where $G = \text{RVIab}(\mathcal{V})$ and $\mathcal{R}_G$ is the same correspondence computed on the kernel: points of $\mathcal{V} \setminus G$ may admit controls that are instantaneously safe while every trajectory must exit later. The obstruction theorems below are *local certificates against* $\mathcal{R}_{\mathcal{V}}$: exhibiting a state at which even the weaker pointwise condition fails certifies exclusion from the kernel, and this is the reading in which the certificates are used. The correspondence is the bridge between the pointwise (tangency) and the global (kernel) description, and it is the object the observation structure acts upon: an observation-based policy can select controls only as a function of the observation record.

2.3 Observation structures, information states, and epistemic kernels

Conventions. Throughout, the plant evolves in continuous time and a policy acts at review times, holding the selected action until the next review; the observation record is generated by the observation map O (the error, hidden-parameter, and delay extensions are noted below). Between reviews the record $y(t) = O(x(t))$ is observed continuously but cannot change the held action; the certificates of Sections 3.2–3.3 and 3.6 use the record only through what it discriminates (observation-equivalence), so they remain valid under any refinement of the within-interval record that does not separate compatible branches. A policy is observation-based if its action depends on the record only through the information set B_t ; under the delay-free set-membership semantics used here the information set is a sufficient statistic, so observation-based and record-based policies coincide (with delays, hidden parameters, or history-dependent constraints the belief must be augmented and sufficiency is not automatic; Section 6.5). The disturbance is chosen adversarially after the control (the Isaacs order of Theorems 4 and 2), an action outside $U(x)$ at a compatible state x is itself a failure (Section 2.4), and constraint violation is the strict inequality $q < 0$.

An **observation structure** $\mathcal{I} = (Y, O)$ is a measurable space Y and an observation map $O : X \rightarrow Y$. The policy observes $y(t) = O(x(t))$. At time t , the **information set** (belief) of a regulator who knows the dynamics, the disturbance class, its own applied controls $(u(s))_{s < t}$, and the record $(y(s))_{s \leq t}$ is

$$\begin{aligned} B_t &= \{\xi(t) : \xi(0) \in B_0, \\ &\quad \dot{\xi}(s) = f(\xi(s), u(s), d(s)), \quad d(s) \in D(\xi(s)), \\ &\quad O(\xi(s)) = y(s) \text{ for all observed } s \leq t\}, \end{aligned}$$

where the applied control history $u(\cdot)$ in the dynamics is the regulator’s own — common to every candidate trajectory — and $\xi(\cdot)$ ranges over candidate trajectories: the set of current states compatible with the record and the applied controls. We assume the standard set-membership semantics (Kurzhanski and Vályi, 1997), so that B_t contains every state consistent with the observations and the applied controls. (With observation error, the equality becomes membership $y(s) \in H(\xi(s), v(s))$; with hidden parameters the belief lives in $X \times \Theta$; with delays it lives in a history space — the extensions are noted as open in Section 6.5.) A policy is **observation-based** if $u(t)$ depends on the record only through B_t ; this is a special case of record-based policies, and the two coincide when the belief is a sufficient statistic for the decision problem, which holds in the delay-free set-membership semantics used throughout.

We write $U^B(B) = \bigcap_{x \in B} U(x)$ for the controls available at every compatible state — the only controls an observation-based policy may select without risking inadmissibility.

Definition 1 (robust epistemic kernel). $B_0 \in \text{ERVIab}_{\mathcal{I}}(\mathcal{V})$ if there exists a nonanticipative observation-based policy π such that, for every compatible initial state $x_0 \in B_0$ and every admissible disturbance realization $d(\cdot)$, the trajectory of $\dot{x} = f(x, u(t), d(t))$ generated from x_0 by π ’s actions under $d(\cdot)$ — the record being the one that $(\pi, d(\cdot))$ generates — remains in $\mathcal{V}$ for all time. (This is the standard exists-strategy-for-all-realizations form. The quantifier order is essential: the record is generated by the realization, so conditioning the admissible realizations on the record would be circular.) Admissible initial beliefs are the compact information sets $B_0 \subseteq X$ generated by the observation structure — the observation fibres $O^{-1}(y)$, $y \in O(X)$, in the constructions of Sections 3.2–3.3 and 3.6. The existential (non-robust) counterpart $\text{EVIab}_{\mathcal{I}}(\mathcal{V})$ is the non-robust contrast class.

We use ERVIab throughout, since the disturbance classes of sustainability problems are adverse by construction. The theorems are stated for this robust notion; they do not transfer a fortiori to the non-robust contrast class $\text{EVIab}_{\mathcal{I}}(\mathcal{V})$, in which there exist an observation-based policy and *some* admissible disturbance realization under which every compatible trajectory remains in $\mathcal{V}$ for all time. Since the robust kernel is contained in the non-robust one, an observation-based policy may exist there that no robust policy survives.

The thesis is now precise: **the epistemic kernel is the greatest recursively viable collection of information states in the sense of the estimation-space reduction cited in Section 5; common-action compatibility is a necessary one-step condition, but not by itself a complete characterization — and the obstruction calculus certifies, mechanism by mechanism, the information states outside it.**

2.4 Notation

We collect the principal symbols used throughout the paper and fix the two structural symbols: the **observation structure** is always $\mathcal{I}$ (calligraphic I) and the **institution** of Section 6.4 is always $\mathcal{J}$ (calligraphic J); no other letter serves either role. The state space is $X \subseteq \mathbb{R}^n$; the control set is U (set-valued $U(x)$ at state x); the disturbance set is D (set-valued $D(x)$). The dynamics are $\dot{x} = f(x, u, d)$. The closed constraint set is $\mathcal{V} \subseteq X$, described by finitely many C^1 constraint functions q_j ($j = 1, \dots, m$). The safe-control correspondence at $x \in \mathcal{V}$ is $\mathcal{R}_{\mathcal{V}}(x)$. The observation structure is $\mathcal{I} = (Y, O)$ with observation map $O : X \rightarrow Y$; the information set (belief) at time t is B_t ; the initial belief is B_0 — admissible initial beliefs are the compact information sets the observation structure generates, the observation fibres $O^{-1}(y)$, $y \in O(X)$, in the constructions. The common admissible action set at belief B is $U^B(B) = \bigcap_{x \in B} U(x)$ — the only controls an observation-based policy may select without risking inadmissibility, under the convention, used from Proposition 5 on, that an action outside $U(x)$ at a compatible state is itself failure; the common safe-action set is $\mathcal{R}_{\mathcal{V}}^B(B) = \bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x)$. The tube-safe action set at review length Δ is $\mathcal{A}_{\text{tube}}(B, \Delta) = \{a : \text{Reach}_{[0, \Delta]}(B, a) \subseteq \mathcal{V}\}$, where $\text{Reach}_{[0, \Delta]}(B, a)$ is the *robust* — all-disturbance, all-solution — reachable set of the system from B under the held action a over $[0, \Delta]$. The first informative observation time is T_{obs} : the first time the observation record discriminates between compatible branches — before it, the branches of Theorem 3’s (H3.1) produce identical records. The adverse-selection correspondences of the exit certificates are $D_{\varepsilon}(x, u) = \{d \in D(x) : D^+q(x; f(x, u, d)) \leq -\varepsilon\}$ (Theorem 4) and $D_{\eta}(x) = \{d \in D(x) : \nabla q_j(x) \cdot f(x, a, d) \leq -\eta/2\}$

(Theorem 2's proof, local). The upper right Dini derivative of q in direction v is $D^+q(x; v) = \limsup_{h \downarrow 0} [q(x + hv) - q(x)]/h$; for q of class C^1 — the class assumed in the exit theorems — $D^+q(x; v) = \nabla q(x) \cdot v$.

The kernels of the hierarchy are $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}})$ (perfect information), $\text{RViab}(\mathcal{V})$ (robust), and $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$ (the robust epistemic kernel — Definition 1, the object of this paper), with $\text{proj}_{\exists}(\mathfrak{B}) = \bigcup_{B \in \mathfrak{B}} B$ the existential projection of a collection $\mathfrak{B}$ of information states. Consistency check: when the observation O is injective the record determines the state, so the epistemic and robust kernels agree, $\text{ERViab}_{\mathcal{I}}(\mathcal{V}) = \text{RViab}(\mathcal{V})$, the informational certificates of Sections 3.2–3.3 and 3.6 reduce to the no-merge case, and only the dynamic obstruction of Section 3.5 can certify exclusion — the definitions degenerate to classical robust viability exactly when observation is perfect. The non-robust counterpart $\text{EViab}_{\mathcal{I}}(\mathcal{V})$ is the non-robust contrast class (Section 2.3). The institutionally restricted kernel is $\text{IRViab}_{\mathcal{I}}(\mathcal{V})$ (Section 6.4). The policy classes are π_{perf} (state feedback), Π_{output} , Π_{state} , and Π_{CE} — the certainty-equivalence class defined in Remark 3: fixed perfect-information regulation laws applied uncorrected to the observation. The policy-specific safety set of Section 4.1 is $\text{Safe}_{\mathcal{I}}^{\Pi}$; the certainly-safe set of Corollary 1 is $\mathcal{Y}_{\text{safe}}$.

Some letters carry site-local meanings. The strip width a of Theorem 4's $\mathcal{S}_a$ is not the held action $a = \pi(B)$ of Theorem 2's proof, nor the pathway vector $a \geq 0$ of Section 5(d); λ is the Farkas multiplier of Theorem 2's checkability certificate and the observer decay rate of Section 5(c); and α is the comparison function of Remark 2 and a multiplier of Section 5(d)'s substitution alternative. The letter K denotes the safe set of Section 4's certification problems, while in Section 5(c) it denotes the kernel itself, with the buffered set K_{ζ} and eroded kernels $K^{-\zeta}$ of Section 5(c). The observation-error map of Section 2.3 is $H(\xi, v)$, whereas the harvest vector of the Supplementary Material is the lowercase h ; and the finite-horizon recursion set $\mathcal{W}_N$ of Section 3.1 is calligraphic, distinct from the Lyapunov function $W = S_1 + S_2$ of the Supplementary Material.

The four correspondences of this section are one skeleton. An observation-based regulator selects a single response for an entire information set — a held action, a feedback segment, or a safe/unsafe verdict — and observation-based viability is the existence of a response that is simultaneously admissible, safe, and recursively viable for every compatible state and disturbance history. Each obstruction certificate of Sections 3 and 4 is a checkable sufficient condition under which one of these response correspondences is empty: $U^B(B)$ (admissibility), $\mathcal{R}_{\mathcal{V}}^B(B)$ (instantaneous safety), $\mathcal{A}_{\text{tube}}(B, \Delta)$ (review-interval safety), or the recursive predecessor of Section 3.1. The mechanisms differ in which correspondence fails, and the obstructions are nested (Proposition 3). The organizing claim of the calculus is the following, stated once and used implicitly by every certificate of Sections 3 and 4.

Proposition 1 (selector principle). If $B \in \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ and π is a policy witnessing viability on B , then at every review time the action $a = \pi(B)$ selected by π is a response that is (i) *admissible*, $a \in U^B(B)$; (ii) *safe*, keeping every compatible trajectory in $\mathcal{V}$ until the next observation — $\text{Reach}_{[0, \Delta]}(B, a) \subseteq \mathcal{V}$ at review length Δ ; and (iii) *recursively viable*, in that every post-observation belief reached at the next review is again in $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$ under the continuation of π . A belief is therefore viable only if each of the response correspondences of this section — $U^B(B)$, $\mathcal{R}_{\mathcal{V}}^B(B)$, $\mathcal{A}_{\text{tube}}(B, \Delta)$, and the recursive predecessor of Section 3.1 — is nonempty along every reachable branch; hence a certificate that any one of them is empty at a reachable belief certifies nonviability. In finite systems the converse is exact (Theorem 1); in continuous time the converse is the backward-recursion completion noted in Section 6.5.

3. The Obstruction Calculus

3.1 Finite-horizon completeness (finite systems)

In the finite setting the calculus is complete: backward recursion over beliefs — the set-membership analogue of the belief-state recursion of partially observable Markov processes (Smallwood and Sondik, 1973) — is sound and complete on any finite horizon, and nonviability is certified by a finite obstruction tree. The certificates of Sections 3.2–3.3 and 3.5–3.6 are sufficient conditions, not a complete characterization of the epistemic kernel.

Fix finite state, action, disturbance, and observation spaces X, A, D, Y , a transition map $x^+ = F(x, a, d)$, an observation map $y = O(x, a, d, x^+)$, a safe set $\mathcal{V} \subseteq X$, and admissible-action sets $U(x) \subseteq A$. For a belief $B \subseteq X$, an action a , and an observation y , the post-belief is

$$\begin{aligned} \text{Post}(B, a, y) &= \{x^+ \in X : \exists x \in B, d \in D(x), \\ &\quad x^+ = F(x, a, d), y = O(x, a, d, x^+)\}, \end{aligned}$$

and y is possible under (B, a) if $\text{Post}(B, a, y) \neq \emptyset$. With $U^B(B) = \bigcap_{x \in B} U(x)$, define the one-step predecessor of a collection $\mathcal{C}$ of beliefs by

$$\begin{aligned} \text{Pre}(\mathcal{C}) &= \{B \subseteq \mathcal{V} : \exists a \in U^B(B) \text{ such that} \\ &\quad \text{Post}(B, a, y) \in \mathcal{C} \text{ for every possible } y\}, \end{aligned}$$

and the finite-horizon kernels by $\mathcal{W}_0 = \{B : B \subseteq \mathcal{V}\}$, $\mathcal{W}_{k+1} = \text{Pre}(\mathcal{W}_k)$. The N -step common safe-action set at a belief B is $\mathcal{A}_N(B) = \{a \in U^B(B) : \text{Post}(B, a, y) \in \mathcal{W}_{N-1} \text{ for every possible } y\}$, so that $B \in \mathcal{W}_N \iff \mathcal{A}_N(B) \neq \emptyset$. The recursion is exact but of worst-case complexity exponential in $|X|$ — the belief space contains up to $2^{|X|}$ beliefs — so finite-checkability is per belief, not a polynomial-time global procedure.

Theorem 1 (finite-horizon soundness and completeness). An initial belief B_0 admits an observation-based policy that keeps every compatible trajectory in $\mathcal{V}$ for N steps if and only if $B_0 \in \mathcal{W}_N$.

Proof sketch. Both directions are inductions on the horizon. Necessity: a policy safe for N steps picks $a \in U^B(B_0)$, and every possible successor belief is safe for $N - 1$ steps, so $B_0 \in \text{Pre}(\mathcal{W}_{N-1})$. Sufficiency: choose the witness action at each step, and every reachable state remains in $\mathcal{V}$ for N steps.

If $B_0 \notin \mathcal{W}_N$, the failure is certified by a finite **obstruction tree**: at each node (B, k) with $k \geq 1$, for every $a \in U^B(B)$ the adversary can choose a possible observation y with $\text{Post}(B, a, y) \notin \mathcal{W}_{k-1}$; the branches are the compatible observation-and-disturbance choices that force violation within N steps. This is the exact finite-horizon counterpart of the sufficient certificates of the following subsections. The recursion reads B_0 as the belief after the initial observation has been assimilated, and the tree alternates regulator and nature turns — the regulator selects an action, the adversary a compatible observation — the standard alternating structure of partial-information games.

A one-step instance. The discrete one-step abstraction of Example 1 — transition $z^+ = z + \theta u$ on the grid $z \in \{-1, 0, 1\}$, preserving the common-action mechanism — is the smallest obstruction tree. With $z \in \{-1, 0, 1\}$, $\theta \in \{-1, +1\}$, $u \in \{-1, +1\}$, transition $z^+ = z + \theta u$, safe set $\mathcal{V} = \{z \geq 0\}$, and observation $O(z, \theta) = z$, the belief $B_0 = \{(0, -1), (0, +1)\}$ has $U^B(B_0) = \{-1, +1\}$. Under $u = +1$ the branch $\theta = -1$ reaches $z^+ = -1$ (observed -1), so $\text{Post}(B_0, +1, -1) = \{(-1, -1)\} \notin \mathcal{W}_0$; under $u = -1$ the branch $\theta = +1$ reaches $z^+ = -1$. Hence $B_0 \notin \mathcal{W}_1$, and the obstruction tree is the root $(B_0, 1)$ with two adversary edges, $\theta = -1$ and $\theta = +1$, each terminating in the violation $z < 0$. This is Theorem 2 (Section 3.2) in the

finite-horizon language; the tree is drawn in the Supplementary Material (Figure S2).

3.2 The instantaneous common-action obstruction

The instantaneous obstruction is the static core of the calculus: it certifies nonviability at a single information state, without a timing or horizon argument.

Theorem 2 (common-action obstruction). Let $B \subseteq \mathcal{V}$ be an information set of the specified observation structure (the safe-control correspondence $\mathcal{R}_{\mathcal{V}}$ is defined on $\mathcal{V}$, so the belief is taken inside it). Suppose the following hypotheses hold.

(H2.1) The selected action is held until the next observation (sampled review: no informative observation arrives before the next action must be selected, and the policy holds the selected action over the review interval — the institutional reading; without a positive dwell time the instantaneous statement below must be replaced by the tube-safety statement below).

(H2.2) Either the **common admissible action set** is empty,

$$U^B(B) = \bigcap_{x \in B} U(x) = \emptyset, \quad (\text{admissibility obstruction})$$

or the **common safe-action set** is empty while common admissibility holds,

$$\mathcal{R}_{\mathcal{V}}^B(B) := \bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x) = \emptyset, \quad (1)$$

with $U^B(B) \neq \emptyset$ (safety obstruction)

(H2.3) **Closed-loop existence of the local adverse selection (safety case).** The disturbance correspondence is lower semicontinuous, or locally constant near the boundary point at which the obstruction fires, so that the local adverse selection of the proof is realized along the compatible trajectory; alternatively, the certificate is read against the convexified (relaxed) inclusion, which preserves the signed drift (for $C^1 q$ the directional derivative is affine in the velocity).

Then, under hypotheses (H2.1)–(H2.3), $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$: every compatible state may be individually robustly viable, and the belief is nevertheless nonviable, because the state-specific safe actions are incompatible.

Proof sketch. If the selected action is inadmissible at some compatible state, the policy fails there outright. Otherwise $a \in U^B(B)$, and the emptiness of the common safe-action set is caused by the boundary: some $\bar{x} \in B \cap \partial\mathcal{V}$ has $a \notin \mathcal{R}_{\mathcal{V}}(\bar{x})$, giving an active constraint j and a disturbance $\bar{d}$ with $\nabla q_j(\bar{x}) \cdot f(\bar{x}, a, \bar{d}) < 0$. A measurable local adverse selection (closed graph of D , continuity of the drift; realized by (H2.3)) then drives q_j strictly downward from $q_j(\bar{x}) = 0$ within the first review period, before any observation can arrive.

Condition (1) is checkable: it requires computing the intersection of the safe-control sets over the information set, which for polyhedral constraint and control sets is a finite linear program. When the intersection is nonempty, the common action is a candidate for viability — Theorem 2 is a *necessary* condition with a constructive witness of failure, not a sufficient condition for success. For polyhedral data the emptiness certificate is a Farkas pair: with safe actions given by finitely many affine inequalities $A_i u \leq b_i$ across compatible states, stacked as $Au \leq b$, the intersection is empty if and only if there exists $\mu \geq 0$ with $\mu^\top A = 0$ and $\mu^\top b < 0$ (Farkas, 1902; Gale, 1960) — a finite, independently checkable certificate of epistemic infeasibility, the control-space analogue of the material-substitution separation certificates in the assessment-separation analysis of Abaee (2026). The certificate is finite when the safe-action constraints over the belief admit a finite representation (finitely

many constraints across the compatible states); for infinite beliefs a finite-witness reduction is not automatic — the Helly-type witness of Proposition 4 supplies one under the convexity hypotheses stated in Section 3.4.

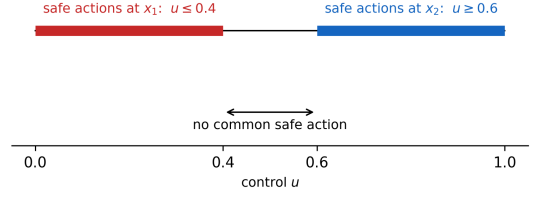


Figure 1: Incompatible safe controls. The states x_1 and x_2 share an observation, but their safe-action sets $[0, 0.4]$ and $[0.6, 1]$ are disjoint, so no common safe action exists (Theorem 2).

The tube-safety form. The instantaneous condition (1) is the $\Delta \downarrow 0$ boundary approximation of the exact one-step obstruction: with the action held over a review interval of length Δ , the exact condition is the emptiness of the tube-safe action set

$$\mathcal{A}_{\text{tube}}(B, \Delta) = \left\{ a : \text{Reach}_{[0, \Delta]}(B, a) \subseteq \mathcal{V} \right\},$$

with $\text{Reach}_{[0, \Delta]}(B, a)$ the robust (all-disturbance) reachable set (the reading specified in Section 2.4), and $\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset$ certifies $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ for every review interval of length Δ and every target collection. The theorems above and below use the instantaneous and timing readings; the tube form is the unifying one-step object.

Proposition 2 (uniform margin gives a tube obstruction at every review length). Let $B \subseteq \mathcal{V}$ be a compact information set with $U^B(B) \neq \emptyset$, f and ∇q_j continuous, and suppose there is $\eta > 0$ such that for every $a \in U^B(B)$ there exist a boundary state $x_a \in B \cap \partial\mathcal{V}$, an active constraint j with $q_j(x_a) = 0$, and a disturbance $d_a \in D(x_a)$ satisfying

$$\nabla q_j(x_a) \cdot f(x_a, a, d_a) \leq -\eta.$$

Under the closed-loop existence convention of Theorem 2, the tube-safe action set is then empty at every review length,

$$\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset \quad \forall \Delta > 0,$$

and hence $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ under sampled review of any length.

Proof sketch. For fixed $a \in U^B(B)$, continuity of the drift and the closed graph of D extend the margin to $-\eta/2$ on a neighbourhood of (x_a, a, d_a) , and a measurable local selection realizes it; the trajectory from x_a under the held action then satisfies $q_j(x(t)) \leq -(\eta/2)t < 0$ for small t , so $a \notin \mathcal{A}_{\text{tube}}(B, \Delta)$ for every $\Delta > 0$. As a was arbitrary, the tube-safe action set is empty.

Proposition 3 (obstruction ladder). Under the standing regularity assumptions and the closed-loop existence convention of Theorem 2, the common response sets of Section 2.4 are nested:

$$\mathcal{A}_{\text{tube}}(B, \Delta) \subseteq \mathcal{R}_{\mathcal{V}}^B(B) \subseteq U^B(B) \quad \forall \Delta > 0,$$

and emptiness descends the ladder:

$$\begin{aligned} U^B(B) = \emptyset &\implies \mathcal{R}_{\mathcal{V}}^B(B) = \emptyset \implies \mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset \\ &\implies B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V}). \end{aligned}$$

Proof sketch. The inclusion $\mathcal{R}_{\mathcal{V}}(x) \subseteq U(x)$ gives the second nesting. For the first, an action outside $\mathcal{R}_{\mathcal{V}}^B(B)$ violates

$\nabla q_j(x) \cdot f(x, a, d) \geq 0$ at some active boundary state, and the corresponding trajectory leaves $\mathcal{V}$ in arbitrarily small time, so the action is not tube-safe.

The obstruction ladder. Each certificate is a sufficient condition for emptiness at one rung:

- *Admissibility obstruction* — $U^B(B) = \emptyset$: no single action is admissible at every compatible state (the base case of Proposition 5).
- *Common safe-action obstruction* — $\mathcal{R}_V^B(B) = \emptyset$ with $U^B(B) \neq \emptyset$ (the safety case of Theorem 2).
- *Tube obstruction* — $\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset$ (the tube-safety form).
- *Recursive obstruction* — in the finite-horizon setting the N -step common safe-action sets $\mathcal{A}_N(B)$ of Section 3.1 complete the ladder $\mathcal{A}_N(B) \subseteq \mathcal{A}_{\text{tube}}(B, \Delta) \subseteq \mathcal{R}_V^B(B) \subseteq U^B(B)$.

The rungs are strict in general: $\mathcal{R}_V^B(B) \neq \emptyset$ does not imply $\mathcal{A}_{\text{tube}}(B, \Delta) \neq \emptyset$ for $\Delta > 0$, since an instantaneously tangent action may still exit in finite time — the gap that Proposition 2 closes under a uniform margin. The dynamic exit certificate of Theorem 4 sits below the ladder: it empties the response set at some compatible state even under full information, which is the strongest form of obstruction and needs no observation-theoretic argument; the ladder, and the position of the exit certificate below it, are drawn in the Supplementary Material (Figure S1).

Example 1 (hidden-mode conflict). Let an unobserved parameter satisfy $\theta \in \{-1, +1\}$, with $\dot{z} = \theta u$, $u \in \{-1, +1\}$, and constraint $z \geq 0$. At $z = 0$: if $\theta = +1$, only $u = +1$ is safe; if $\theta = -1$, only $u = -1$ is safe. Both compatible states are individually robustly viable under full information (each admits its safe control), but the information set $B = \{(0, +1), (0, -1)\}$ satisfies $\mathcal{R}_V^B(B) = \{+1\} \cap \{-1\} = \emptyset$, so $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$. This is a purely informational failure: no stochasticity and no estimation-quality argument is involved. The example is the two-action skeleton of every “the stock is either recovering or collapsing, and the safe policy differs between the two” situation.

3.3 The delayed-information obstruction

Theorem 3 (delayed-information obstruction). Let B_0 be an initial information set. Suppose the following hypotheses hold.

(H3.1) No informative observation arrives before time $T_{\text{obs}} > 0$ — formally, there are branches that are **observation-equivalent** on $[0, T_{\text{obs}})$: they produce identical observation records throughout that interval, so the policy cannot distinguish them before T_{obs} ; between observations the information set evolves by the set-membership semantics of Section 2.3.

(H3.2) There exist a constraint function q of class C^1 (the class of Theorem 4’s proof, Section 3.5) with $\mathcal{V} \subseteq \{q \geq 0\}$ and a constant $\varepsilon > 0$ such that, **for every implementable blind-window control** — every measurable $u(\cdot)$ on $[0, T_{\text{obs}})$ that is admissible at every compatible state throughout the blind window, $u(t) \in U^B(B_t)$ with B_t the set-membership belief propagated under $u(\cdot)$ — which is exactly the class of controls that an observation-based policy can realize before the first informative observation, where its actions are a fixed function of time — there are a compatible initial state $x^* \in B_0$ attaining $q(x^*) = \inf_{x \in B_0} q(x)$ (with B_0 compact and q lower semicontinuous; otherwise replace the infimum by $\inf q + \delta$ for an arbitrary δ -minimizer, let $\delta \downarrow 0$, and read the timing bound with $\inf q + \delta$) and an admissible disturbance realization such that, along the realized trajectory from x^* under that open-loop control, the record remains compatible and

$$D^+ q(x(t); f(x(t), u(t), d(t))) \leq -\varepsilon \quad \text{while } q(x(t)) > 0, \quad (2)$$

(H3.3) The timing bound holds:

$$T_{\text{obs}} > \frac{\inf_{x \in B_0} q(x)}{\varepsilon}. \quad (3)$$

Then, under hypotheses (H3.1)–(H3.3), $B_0 \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$: information may be accurate but arrive too late.

Proof sketch. By (H3.1) the policy’s actions on $[0, T_{\text{obs}})$ form one fixed open-loop control; if it is ever inadmissible at a compatible state the policy fails outright, otherwise (H3.2), applied to this implementable control, supplies a compatible state x^* with $q(x^*) = \inf_{x \in B_0} q$ and an admissible disturbance realizing the drift (2); integrating (2) gives $q(x(t)) \leq \inf_{x \in B_0} q - \varepsilon t$, so the violation $q < 0$ occurs by time $\inf_{x \in B_0} q / \varepsilon$, which (3) places strictly before T_{obs} — before any informative observation, so the policy cannot react in time.

The hypothesis (2) is the set-membership form of the drift condition, quantified over open-loop controls on the blind window — a condition on the data of the problem, not on the policies: for every such control there is a compatible state and disturbance enforcing the drift; the adversary selects the true state and the disturbance, and the observations cannot expose the selection in time. It is this open-loop form that gives the timing bound (3) its force — (H3.1) converts the policy’s actions into one open-loop control, (H3.2) supplies the drift against it, and (3) then makes the obstruction epistemic rather than dynamic. The hypothesis is checkable in its special cases (constant observations; hold-until- T_{obs} policies) and is a template to be instantiated in the general case. Condition (3) is the **timing bound**: it states that the observation arrives *after* the enforced exit time — the cause of the obstruction; inverted, it reads as the design requirement that the first informative observation precede the enforced exit time $\inf q / \varepsilon$. Theorem 3 complements Theorem 2 quantitatively: the common-action obstruction is a **formal zero-margin counterpart** of the delayed tube-safety obstruction, and the two certificates remain distinct — the one a local action-feasibility condition, the other a finite-time survival-value condition — so the pair is not related as a $T_{\text{obs}} \rightarrow \infty$ limit. It is the obstruction that governs discrete-review governance: a review interval longer than the worst-case exit time is an information structure in which the exit is inevitable.

Remark 1 (threshold form of the timing obstruction). The delayed-information obstruction has a sharp threshold. Let $\sigma^*(B_0)$ be the guaranteed blind-window survival time: the largest duration such that some pre-observation policy keeps every observation-equivalent branch in $\mathcal{V}$ until then,

$$\sigma^*(B_0) = \sup_{u(\cdot)} \inf_{x_0 \in B_0, d(\cdot)} \tau(x_0, u, d),$$

with the infimum over branches that remain observation-equivalent on $[0, T_{\text{obs}})$ and τ the first violation time under the convention of Theorem 3. Then

$$\sigma^*(B_0) < T_{\text{obs}} \implies B_0 \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V}),$$

because before T_{obs} a policy cannot condition its action on which branch realizes. Condition (3) certifies this threshold: under (H3.1)–(H3.2) every branch exits by time $\inf_{x \in B_0} q(x) / \varepsilon$, hence $\sigma^*(B_0) \leq \inf_{x \in B_0} q(x) / \varepsilon$, and (3) states that this upper bound lies below T_{obs} ; the threshold itself may be strictly smaller, in which case nonviability holds even when (3) fails. The threshold is the blind-window response correspondence of the timing layer: its emptiness is the timing certificate, the delayed-information analogue of the tube and common-action correspondences.

Checkability of the timing certificate. The drift hypothesis (3) is finitely checkable whenever the blind-window

control class is finite: if B_0 is finite and the admissible blind-window controls form a finite family (hold-until- T_{obs} controls with u in a finite set, or piecewise-constant controls on a finite partition), the certificate reduces to the finite check that every candidate control enforces exit before T_{obs} , each worst-case exit time computed by branchwise integration or by the recursion of Section 3.1. The hidden-regime instance of Section 8 is this check in closed form: the worst branch declines at unit rate, giving $\sigma^*(B_0) = z_0 - 1$, and the certificate fires exactly when $T_{\text{obs}} > z_0 - 1$. For continuous control classes the check is a template rather than a finite procedure; closing that gap is part of Open Problem 1.

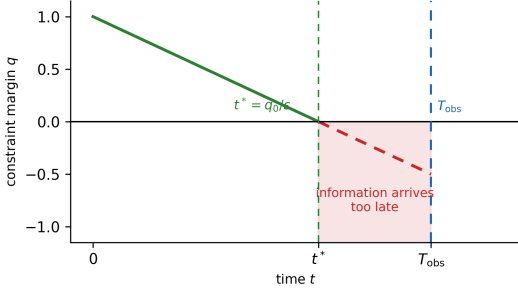


Figure 2: Delayed information. The constraint margin q reaches zero at $t^* = q_0/\varepsilon$, before the first informative observation at T_{obs} : the information is accurate but arrives too late (Theorem 3).

3.4 The sparse finite witness

Proposition 4 (sparse common-action witness). Let $B \subseteq \mathcal{V}$ be an information set, $U(x) \subseteq U$ with $U \subseteq \mathbb{R}^m$ compact and convex and $U(x)$ convex and closed, let the dynamics be affine in the control, $f(x, u, d) = f_0(x, d) + f_u(x, d)u$, with $D(x)$ compact, and let $\mathcal{V}$ have C^1 constraint functions q_j . Then each safe-control set $\mathcal{R}_{\mathcal{V}}(x)$ is a compact convex subset of U , and the safety-case common-action obstruction $\bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x) = \emptyset$ holds if and only if it is witnessed by at most $m+1$ compatible states: there exist $x_1, \dots, x_{m+1} \in B$ with $\bigcap_{i=1}^{m+1} \mathcal{R}_{\mathcal{V}}(x_i) = \emptyset$. The admissibility analogue (with $\mathcal{R}_{\mathcal{V}}(x)$ replaced by $U(x)$) holds under the same convexity assumptions.

Proof. For fixed (x, j, d) the inequality $\nabla q_j(x) \cdot f(x, u, d) \geq 0$ is affine in u , so $\mathcal{R}_{\mathcal{V}}(x)$, an intersection of closed halfspaces (over active j and $d \in D(x)$) with the closed convex $U(x)$, is closed and convex; contained in the compact U , it is compact. If the intersection over all $x \in B$ is empty, then by compactness some finite subfamily of $\{\mathcal{R}_{\mathcal{V}}(x)\}_{x \in B}$ already has empty intersection, and by the finite Helly theorem at most $m+1$ members suffice; the converse direction is immediate. $\square$

A worked certificate (two floors, one scalar action). Two floors demand incompatible extraction rates on a single control $u \in [0, 1]$. At the compatible state x_1 (floor 1 active) the safe controls are $u \leq 0.4$; at x_2 (floor 2 active) they are $u \geq 0.6$. The observation merges x_1 and x_2 , so a common safe action must satisfy both. Writing the constraints as $Au \leq b$ with $A = (1, -1)^\top$ and $b = (0.4, -0.6)^\top$, the stacked system $u \leq 0.4$, $-u \leq -0.6$ is infeasible; the Farkas certificate is $\lambda = (1/2, 1/2)$ (normalized, $\|\lambda\|_1 = 1$), for which $\lambda^\top A = 0$ and $\lambda^\top b = -0.1 < 0$. Under the normalization $\|\lambda\|_1 = 1$, the magnitude $|\lambda^\top b| = 0.1$ is the infeasibility margin of the stacked system, shrinking to zero exactly as the two safe-action sets come to touch. The multiplier exhibits the contradiction as a convex combination of the two floor constraints: no common extraction rate respects both floors, and any observation that separates x_1 from x_2 restores one-step viability (Figure 1).

With $m = 1$, the two incompatible states are exactly the $m+1$ -state witness of Proposition 4.

Scope of the convexity hypotheses. Convexity is essential to the witness: without it the bound fails even in a single control dimension. Take $U = \mathbb{R}$ ($m = 1$) and three nonconvex safe-action sets $\mathcal{R}_1 = \{0, 1\}$, $\mathcal{R}_2 = \{0, 2\}$, $\mathcal{R}_3 = \{1, 2\}$. Each pair intersects ($\mathcal{R}_1 \cap \mathcal{R}_2 = \{0\}$, $\mathcal{R}_1 \cap \mathcal{R}_3 = \{1\}$, $\mathcal{R}_2 \cap \mathcal{R}_3 = \{2\}$), yet the three-way intersection is empty, so no $(m+1) = 2$ -state witness exists. The convexity hypotheses of Proposition 4 exclude exactly this failure mode.

3.5 The finite-time exit certificate

The base obstruction operates before any observation-theoretic argument: it certifies membership in the full-information “shadow” — the complement of the viability kernel (Aubin, 2001; Aubin and Catté, 2002). It is a drift condition under which *no* control — of any information structure — can be viable. Put in management terms, when the dynamics themselves force exit before any control can react, no monitoring design can rescue viability, and the certification problem is closed without observation-theoretic reasoning.

Theorem 4 (finite-time exit certificate). Let $q : X \rightarrow \mathbb{R}$ be a constraint function of class C^1 — the class assumed throughout the exit theorems — with $\mathcal{V} \subseteq \{q \geq 0\}$, and suppose the following hypotheses hold.

(H5.1) There exist constants $a > 0$ and $\varepsilon > 0$ such that on the strip $\mathcal{S}_a = \{x : 0 \leq q(x) \leq a\}$,

$$\sup_{u \in U(x)} \inf_{d \in D(x)} D^+ q(x; f(x, u, d)) \leq -\varepsilon \quad \forall x \in \mathcal{S}_a, \quad (4)$$

where $D^+ q(x; v) = \limsup_{h \downarrow 0} [q(x + hv) - q(x)]/h$ is the upper right Dini derivative of q in direction v (for C^1 q , $D^+ q(x; v) = \nabla q(x) \cdot v$ — the form the proof uses).

(H5.2) **Closed-loop existence of the adverse realization.** The disturbance correspondence is lower semicontinuous, or constant, on the strip, so that the state-dependent adverse selection constructed in the proof is realized by an admissible control–disturbance pair for which the Carathéodory existence theorem applies. (Alternatively, without (H5.2), the certificate is read against the convexified (relaxed) inclusion $\dot{x} \in \text{co } f(x, u(t), D_\varepsilon(x, u(t)))$, which admits solutions under Marchaud-type conditions and preserves the drift inequality because, for C^1 q , the directional derivative $\nabla q(x) \cdot v$ is affine in the velocity; closure and relaxation are the standard tools for this step.)

Then, under hypotheses (H5.1)–(H5.2), for every admissible control and every initial state $x_0 \in \mathcal{S}_a$ there exists an admissible disturbance realization under which the trajectory attains $q(x(t)) < 0$ — hence leaves $\{q \geq 0\} \supseteq \mathcal{V}$ — by every time strictly larger than $q(x_0)/\varepsilon$: $\inf\{t : q(x(t)) < 0\} \leq q(x_0)/\varepsilon$, in particular within time at most a/ε (since $q(x_0) \leq a$). (Convention, used in both exit theorems: *violation* is the strict inequality $q < 0$; at $q = 0$ the state may still lie on $\partial \mathcal{V} \subseteq \mathcal{V}$.)

Proof sketch. Fix an admissible control. By (4), the adverse-selection correspondence $D_\varepsilon(x, u)$ has nonempty values on the strip; its closed graph and compact values (Section 2.1) yield a measurable selection $d(\cdot)$, realized in closed loop by (H5.2). The fundamental theorem of calculus integrates (4) along the resulting trajectory to $q(x(t)) \leq q(x_0) - \varepsilon t$, so q attains the strict violation $q < 0$ by time $q(x_0)/\varepsilon \leq a/\varepsilon$.

Remark 2 (comparison-function form). Hypothesis (H5.1) uses a constant drift margin ε . The certificate extends to state-dependent decline: if there is a continuous $\alpha : (0, a] \rightarrow (0, \infty)$ with

$$\sup_{u \in U(x)} \inf_{d \in D(x)} D^+ q(x; f(x, u, d)) \leq -\alpha(q(x)) \quad \forall x \in \mathcal{S}_a,$$

then the exit-time bound sharpens to

$$t_{\text{exit}} \leq \int_0^{q(x_0)} \frac{ds}{\alpha(s)},$$

whenever the integral is finite. The proof replaces the linear comparison $q(x(t)) \leq q(x_0) - \varepsilon t$ with the solution r of $\dot{r} = -\alpha(r)$, $r(0) = q(x_0)$, by the standard comparison argument. If $\alpha(0) = 0$ and the integral diverges, the conclusion weakens to asymptotic approach rather than finite-time exit; the constant-margin case $\alpha \equiv \varepsilon$ is the sharp finite-time instance.

Two readings of (4) matter. First, it is an **Isaacs-type condition**: the disturbance chooses after the control, so the certificate asserts that the disturbance can *enforce* exit against any policy. Second, (4) is an outward-drift certificate *dual in spirit*, but *not logically complementary*, to an inward barrier condition. The pointwise robust barrier reading is $\sup_u \inf_d D^+ q \geq 0$ — some control keeps the drift inward against every disturbance — and the two conditions leave an unresolved intermediate region in which neither holds (Section 6.2). Theorem 4 is the “unsafety” side of that duality, and it is unconditional: it needs no observation argument because it defeats all controls, including clairvoyant ones.

3.6 Epistemic emptiness by admissibility: a minimal construction

The admissibility obstruction applies to systems in which every state is individually viable under full information and the kernel empties only under the observation structure. The minimal construction of this section empties the kernel through *admissibility* — a state-dependent control set whose common action is forced and unsafe at the boundary; the common-action obstruction of Section 3.2 and Example 1’s hidden-parameter conflict are the informational cases proper, merging states with incompatible safe controls. In ecological terms, each stock is manageable on its own; the failure arises because the indicator cannot tell the manager which stock is present.

Proposition 5 (epistemic emptiness by admissibility — a minimal construction). There exists a system with $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}}) = \mathcal{V} \neq \emptyset$ such that, for a non-injective observation map O , every admissible initial information state — the observation fibres $B_0 = O^{-1}(O(S_0))$, $S_0 \in \mathcal{V}$ — lies outside $\text{ERViab}_T(\mathcal{V})$: the epistemic kernel over the fibre-induced initial beliefs is empty. The mechanism is *admissibility* rather than a hidden mode: the state-dependent control set of the construction leaves a single common action at the merged belief — one that is outward-drifting at the boundary state — so the emptiness is the common-action obstruction (Theorem 2’s safety case) at minimal size, produced by the control-set primitive rather than by information; the hidden-mode instance is Example 1.

Proof sketch (construction). Take $X = [1, 2]$, $\dot{S} = u - r(S)$ with r continuous and injective, $U(S) = \{0, r(S)\}$, $\mathcal{V} = [1, 2]$, and the constant observation $O \equiv 0$. Under full information $u = r(S)$ gives $\dot{S} = 0$, so every state is viable. Under $O \equiv 0$ the common control set over the fibre is $\{0\}$ — injectivity of r rules out a common $r(S)$ — so the policy must hold $u = 0$, and $\dot{S} = -r(S) < 0$ drives every compatible trajectory out of $\mathcal{V}$ in finite time.

The construction isolates an admissibility mechanism: the observation merges states whose *admissible* controls differ — the merged belief admits a common admissible control (the singleton $\{0\}$) but no common *safe* control, because that singleton is outward-drifting at the boundary state $S = 1$. The “inadmissible action = failure” convention is used explicitly: an action outside $U(x)$ at a compatible state x is itself failure, and the state-dependent control set $U(S) = \{0, r(S)\}$ is a modelling

primitive carrying that convention (Theorem 2’s admissibility case uses the same convention, stated there). The example also shows the empty-kernel phenomenon at its minimal size — two control values, one scalar state — as the minimal instance of Theorem 2 rather than a mechanism of equal rank; with a constant control set $[0, \bar{u}]$ in the same plant the fibre is epistemically viable (a constant control $u = r(m)$, $m \in (1, 2)$, drives every state of the fibre to the rest point $m \in (1, 2)$), so the construction is an admissibility mechanism rather than a hidden-mode mechanism. Theorem 2 states the general obstruction, of which this construction is the minimal instance.

4. Certification Limits

4.1 Exact certification and the fibre criterion

The obstruction theorems of Section 3 concern *policies*. Sustainability governance also uses *certificates*: verdicts — safe or unsafe — computed from observations alone, without reference to a policy. The question is when such verdicts can be exact. For a manager operating under partial ecological information, the question is whether an indicator reading alone can ever settle whether the system is on the safe side of a specified floor.

Definition 2 (exact certifier). For an admissible domain $Z \subseteq X$ and a safe set $K \subseteq Z$, an **exact certifier** based on the observation map $O : Z \rightarrow Y$ is a function $C : Y \rightarrow \{0, 1\}$ such that

$$C(O(z)) = 1 \iff z \in K \quad \text{for every } z \in Z.$$

Proposition 6 (observation-fibre criterion). An exact certifier based on O exists if and only if membership in K is constant on every observation fibre:

$$O(z_1) = O(z_2) \implies [z_1 \in K \iff z_2 \in K] \quad \forall z_1, z_2 \in Z, \quad (5)$$

equivalently, $K = O^{-1}(O(K))$ on the admissible domain.

Proof sketch. If C exists and $O(z_1) = O(z_2)$, then $\mathbf{1}_K(z_1) = C(O(z_1)) = C(O(z_2)) = \mathbf{1}_K(z_2)$, so membership is fibre-constant. Conversely, (5) makes $C(y) = \mathbf{1}_K(z)$ for any z with $O(z) = y$ well defined and exact.

The fibre criterion concerns static classification — whether membership in a safe set factors through the observation — and applies verbatim to three distinct certification problems, which should not be conflated: constraint certification ($K = \mathcal{V}$), robust viability certification ($K = \text{RViab}(\mathcal{V})$), and policy-specific safety certification ($K = \text{Safe}_T^\Pi$). The same theorem governs all three, but its governance meaning differs. The criterion is the zero-step, label-valued analogue of the common-action test: an observation-based verdict assigns a single label to an entire fibre, exactly as an observation-based action assigns a single action to an entire belief, so both are selector problems. They remain distinct — a fibre may fail exact certification while a policy is still viable, and exact static certification does not imply dynamic viability; hence certification is treated as a layer separate from viability throughout.

Corollary 1 (safety-crossing fibres and the certainly-safe set). If two admissible states share an observation and lie on opposite sides of the *same* component safety constraint — with the safe set of that certification problem read as the single floor $K = \{q_j \geq 0\}$; or, in the general form of the hypothesis, one of the two states lies in K and the other outside K — no exact observation-only certificate exists. (Crossing one floor of an intersection $K = \bigcap_j \{q_j \geq 0\}$ does not by itself put the pair on opposite sides of the intersection — the state satisfying floor j may violate floor i — hence the single-floor scoping.) The

largest set of observations that can soundly be labeled safe without further information is the certainly-safe set

$$\mathcal{Y}_{\text{safe}} = \{y \in O(Z) : O^{-1}(y) \subseteq K\}.$$

Proof sketch. A fibre containing one state in K and one outside violates (5), so Proposition 6 denies the certifier. On $\mathcal{Y}_{\text{safe}}$ every compatible state is safe, so the constant verdict “safe” is sound; outside it some compatible state is unsafe, so no sound “safe” verdict exists.

The fibre criterion has a direct governance reading. A composite sustainability index is an observation map from the state of a socio-ecological system to a scalar; a floor is a safe set. Corollary 1 then states: an index whose fibres cross a floor boundary — whose value cannot distinguish a state violating the floor from one satisfying it — cannot support an exact certification of that floor, however carefully the index is thresholded. The structural requirement is not that each floor be measured separately, but that the observation *separate every admissible pair lying on opposite sides of a specified floor*; direct per-floor measurement is one sufficient design, not the unique one, and any statistic refining the floor’s safe/unsafe partition certifies equally well. The certainly-safe set is the sound relaxation: it is the set of observation values where the index may certify safety; outside it, the index must fall silent.

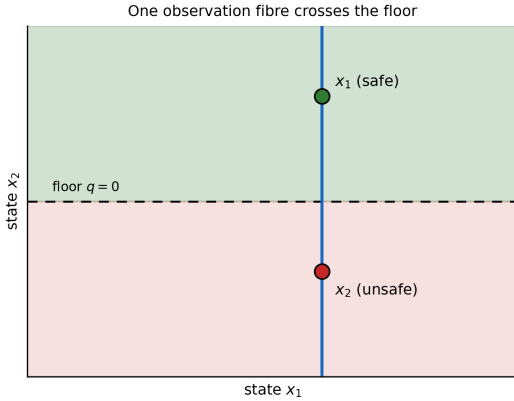


Figure 3: An observation fibre crossing a floor boundary. The observation $O(x) = x_1$ cannot separate the safe state $(0.6, 0.8)$ from the unsafe state $(0.6, 0.3)$ on the same fibre, so no exact observation-only certifier exists (Proposition 6).

4.2 The certainty-equivalence trap

The obstruction of Proposition 5 relies on a non-injective observation — information lost. The same emptying occurs with a fully *injective* observation for a single policy: the canonical certainty-equivalence controller that applies a perfect-information law to the uncorrected observation.

Remark 3 (the certainty-equivalence trap). Consider $\dot{S} = u - g(S)$ with $u \in [0, \bar{u}]$, g strictly increasing on $[S_{\min}, S^* + b]$, $g(0) = 0$, and $\mathcal{V} = [S_{\min}, S^*]$. Assume the range condition $g(S + b) \leq \bar{u}$ for every $S \in [S_{\min}, S^*]$, so that the controller below is admissible. Under perfect information the zeroing feedback $u(t) = g(S(t))$ gives $\dot{S} = 0$, so every $S_0 \in \mathcal{V}$ is viable and $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}}) = \mathcal{V} \neq \emptyset$. Now take the injective, biased observation $\hat{S} = S + b$ with $b > 0$, and the canonical certainty-equivalence controller: the perfect-information zeroing law applied directly to the uncorrected observation, $u = g(\hat{S}) = g(S + b)$. Then

$$\dot{S} = g(S + b) - g(S) > 0 \quad \forall S,$$

and since g is strictly increasing on the compact interval $[S_{\min}, S^* + b]$, $\dot{S}$ is bounded below by a positive constant there;

hence S strictly increases and exits above S^* in finite time from every $S_0 \in \mathcal{V}$: the kernel of this single controller is empty. The bias-corrected controller $u = g(\hat{S} - b) = g(S)$ uses the observation map’s structure, restores $\dot{S} = 0$, and keeps the kernel nonempty.

The mechanism is a *failure of one policy to use the observation map’s structure*, not a loss of information: for a known invertible observation the output-feedback class is as powerful as the state-feedback class. The phenomenon is the viability-theoretic analogue of Witsenhausen’s counterexample in stochastic control (Witsenhausen, 1968): the information pattern — not the dynamics and not the constraint — defeats the uncorrected controller. The remark is policy-specific; it does not assert that the entire certainty-equivalence class empties the kernel.

5. The Sufficiency Landscape

For contrast, we record the sufficiency results against which the obstruction calculus is defined; the proofs are in the cited sources.

(a) Veliov’s output-feedback condition. Veliov (1993) gives a sufficient condition for the existence of an output-feedback regulation map under incomplete and inexact measurement, reducing under perfect measurement to the classical viability condition (Haddad, 1981). Veliov’s theorem certifies existence; the obstruction calculus certifies nonexistence; a problem satisfying neither requires a stronger observer or a finer observation structure.

(b) The estimation-tube programme. Cardaliaguet, Quincampoix, and Saint-Pierre (2007) pass from imperfect information in the measurement space to perfect information in an estimation space of sets of compatible states, with equality of value functions and a Dini-derivative characterization. Under the set-membership semantics the estimation-space solution propagates the belief under a single control signal, so its controls are common controls; the common-action obstruction (Theorem 2) is the certificate that no jointly admissible selection exists at the information state in question (Section 6.3).

(c) Observer-and-buffer transfer. If a full-information feedback exists with a strict inward margin and an observer supplies exponentially convergent estimates, output feedback of the estimate preserves safety on a buffered subset (observer-to-viability transfer with safety buffer; Sontag, 1998), and eroded kernels remain invariant under bounded estimation and implementation errors (erosion absorbs the error). Under margins and convergence, output feedback can work; Sections 3–4 state when it cannot.

(d) The linear substitution alternative. For a finite linear model in which a resource-typed system must meet a demand vector through specified substitution pathways, exactly one of the following holds (Farkas, 1902; Gale, 1960): (i) there exists a non-negative pathway vector $a \geq 0$ satisfying the linear substitution constraints; or (ii) there exist multipliers $\alpha, \beta, \gamma \geq 0$ with

$$\begin{aligned} \alpha^\top R + \beta^\top E - \gamma^\top Q &\geq 0 \quad \text{componentwise,} \\ \gamma^\top s^{\text{req}} &> \alpha^\top x + \beta^\top e. \end{aligned}$$

The second statement is a separation certificate that the pathways cannot meet demand within the typed resource and capacity bounds — the feasibility-side complement of the observation-side obstructions of Sections 3–4.

6. Discussion

6.1 The shape of the calculus

The six mechanisms form a nonexhaustive taxonomy of information-theoretic failure. Table 1 collects the certificates, the response correspondence each one empties, and the design consequence each one licenses.

The exit certificate (Theorem 4) is the dynamic obstruction: it defeats all controls, so it needs no observation argument. The epistemic-emptiness construction (Proposition 5) exhibits the mechanism at minimal size — one scalar state, two admissible controls, one constant observation — in the admissibility form of Section 3.6. The common-action obstruction (Theorem 2) is the static obstruction: it defeats all policies at a single information state. The delayed-information obstruction (Theorem 3) interpolates between them in time. The fibre criterion (Proposition 6) is the certification obstruction: it limits not policies but verdicts. The certainty-equivalence trap (Remark 3) limits a single uncorrected policy. Finite-horizon completeness (Theorem 1) supplies the exact finite-setting counterpart of these sufficient certificates.

A governance failure of observation can be diagnosed by which certificate applies. If the exit certificate holds, no monitoring design helps — the dynamics are adverse under every control. If only the common-action obstruction holds, the remedy is a finer observation structure (an informative measurement distinguishing the incompatible states). If the delayed-information obstruction holds with a slack timing bound, the remedy is a shorter review interval or a faster indicator. If the fibre criterion fails, the remedy is an observation that separates the sides of the floor rather than a better threshold on the aggregate.

6.2 Relation to barrier certificates

Barrier certificates certify safety from a scalar function whose zero level set separates the unsafe region from all trajectories (Prajna and Jadbabaie, 2004; Prajna, Jadbabaie, and Pappas, 2007); the converse, under convex-duality conditions on density functions, is Prajna and Rantzer (2005), and necessary-and-sufficient hybrid characterizations are Maghenem and Sanfelice (2019). Theorem 4 is the unsafety complement of that converse, not its observation-theoretic counterpart: a Dini-drift certificate of unsafety needing no observation structure, constructive in the enforcing disturbance and exit time. The remaining certificates (Proposition 5, Theorem 2, Theorem 3, Proposition 6) act at the level of the information set itself: barrier constructions presuppose a state — or a specified state estimate — for feedback, whereas the common-action and timing obstructions certify the emptiness of the common safe-action set at a given information state, which barrier methods do not address.

6.3 Relation to the estimation-tube programme

The estimation-tube programme shows that imperfect measurement can be absorbed into set-valued dynamics on an estimation space with no loss in value (Cardaliaguet, Quincampoix, and Saint-Pierre, 2007); the present theorems delimit its reach rather than contradict it. The reduction preserves values, and its belief-state controls are common controls; the obstruction calculus is the local certificate layer of that programme — the common-action obstruction certifies that the belief-state predecessor is empty, the timing bound quantifies when it fails for reasons of information arrival, and the fibre criterion states when no observation-based verdict can be exact. Belief-space viability answers *whether*; the obstruction calculus answers *why not, and what to change*. A single worked system carrying the

three methods side by side — barrier certificate, estimation-tube reduction, and obstruction certificate — is given in the Supplementary Material (S2).

6.4 Consequences for monitoring and indicator design

Five consequences follow directly from the certificates. Each translates a formal impossibility into a concrete design specification for monitoring systems under partial ecological information.

Timing. Theorem 3 gives the minimal monitoring frequency in closed form: the first informative observation must precede the enforced exit time $\inf q/\varepsilon$. A review interval longer than the worst-case time from the most-constrained compatible state — the minimum- q state, which has the least constraint slack — to constraint violation is an information structure under which the violation is undetectable in time, whatever the review then concludes. The bound is computable from the drift certificate (2), which a robustness analysis computes in any case.

Coarseness. Theorem 2 states that an observation structure merging two compatible states with incompatible safe controls is nonviable *at that information state*, even though both states are viable in isolation. The design consequence is that the observation must separate safety classes, not states: an indicator is exposed to Theorem 2’s obstruction when some observation fibre crosses the safe-control partition of the state space — merging states whose safe controls differ is what the theorem certifies, and keeping every fibre within a single class is what removes that exposure; refinement is never harmful (a finer observation merges nothing the coarser one did not), though Theorem 2 claims no benefit beyond the removal, and other mechanisms may still certify nonviability on a finer partition.

Aggregation. Proposition 6 and Corollary 1 apply verbatim to composite indices, which are observation maps from the system state to a scalar. An index whose fibres cross a floor boundary cannot support exact floor certification; the certainly-safe set is the domain in which an index-based verdict is sound. Where separately-binding floors matter, exact certification requires an observation that separates every admissible pair on opposite sides of a floor (per-floor measurement is one sufficient design), a formal complement to the thesis, argued informally since Martinez-Alier, Munda, and O’Neill (1998), that weakly comparable values cannot be fused into one metric without loss. The requirement is sharpest where the floor constrains the productive base rather than its output: an aggregate of services can read at full level while the base regenerating them is reduced beneath it, and only an observation reaching the base itself can detect the discrepancy.

Bias. Remark 3 shows that the loss need not be in the measurement: a biased indicator with an uncorrected certainty-equivalence policy empties the epistemic kernel of a system whose perfect-information kernel is nonempty. Monitoring therefore has two obligations — bounding the observation error and supplying the correction — and the second is the one governance structures can omit: the indicator is published, the correction is not applied, and the kernel empties although the information was sufficient.

Institutions. The hierarchy of Section 2.1 records that epistemic contraction is one of three causes of kernel shrinkage, the others being disturbances and institutional restriction (authority, enforcement, allocation). The epistemic-institutional kernel $\text{IRVib}_{\mathcal{J}}(\mathcal{V})$ (Section 2.1) combines them: an institution restricted in what it may observe *and* in what it may command inherits both contractions. The obstruction calculus characterizes the first; its certificates remain valid, and typically tighten, when the institution’s command set is further restricted, since

Certificate	Emptied response correspondence	Design consequence (Section 6.4)
finite-time exit (Theorem 4)	the response set of some compatible state, even under full information	none — dynamics, not information
admissibility (Proposition 5)	$U^B(B) = \emptyset$	enlarge the command set
common action and tube (Theorem 2, Proposition 2)	$\mathcal{R}_V^B(B) = \emptyset$, $\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset$	a separating observation; a shorter review interval
delayed information (Theorem 3, Remark 1)	$\sigma^*(B_0) < T_{\text{obs}}$	an earlier informative observation
fibre certification (Proposition 6, Corollary 1)	$K \neq O^{-1}(O(K))$	a refined index; per-floor measurement
certainty equivalence (Remark 3)	kernel of the uncorrected controller $u = g(\hat{S})$ is empty	correct the known bias
finite-horizon recursion (Theorem 1)	$B_0 \notin \mathcal{W}_N$	exact characterization (Section 3.1)

Table 1: The obstruction certificates, the response correspondence each one empties, and the design consequence each one licenses. The first six are sound sufficient conditions for nonviability; the last is the exact finite-horizon characterization.

restriction of the action correspondence can only empty kernels, never create them (one-sided monotonicity of viability under correspondence restriction).

Proposition 7 (monotonicity of the epistemic kernel). The robust epistemic kernel is monotone in its constituents: enlarging the action sets, reducing the disturbance sets, refining the information structure, or enlarging the policy class never shrinks the kernel. Writing ERViab^* for the kernel with the indicated constituent, $U_1 \subseteq U_2$, $D_1 \subseteq D_2$, $\Pi_1 \subseteq \Pi_2$, and “ $\mathcal{I}_1$ refines $\mathcal{I}_2$ ” imply respectively

$$\begin{aligned} \text{ERViab}^{U_1} &\subseteq \text{ERViab}^{U_2}, & \text{ERViab}^{D_2} &\subseteq \text{ERViab}^{D_1}, \\ \text{ERViab}^{\Pi_1} &\subseteq \text{ERViab}^{\Pi_2}, & \text{ERViab}^{\mathcal{I}_2} &\subseteq \text{ERViab}^{\mathcal{I}_1}. \end{aligned}$$

Proof sketch. A policy viable under the smaller action sets, larger disturbance sets, smaller policy class, or coarser information structure remains admissible and nonanticipative under the corresponding enlargement or refinement, while the adversary’s options are unchanged or reduced; the same policy witnesses viability.

6.5 Limitations

The certificates are sound but not complete; several extensions are noted as open.

- (i) The continuous-time certificates of Sections 3.2–3.3 and 3.5–3.6 are sufficient conditions for nonviability, not necessary-and-sufficient characterizations of the epistemic kernel (backward recursion is complete in the finite-horizon finite-system setting, Section 3.1); the middle ground (neither a Veliov-type condition nor an obstruction certificate) is open. In continuous settings completeness would additionally require a measurable-selection theory beyond the explicit selections the certificates exhibit; completeness is therefore claimed only where backward recursion is exact (Section 3.1).
- (ii) The timing obstruction of Theorem 3 (and its threshold form, Remark 1) certifies failure by exit *before* the first informative observation. It is blind to the complementary mode of nonviability — *insufficient post-observation recourse* — in which a common blind-window control keeps every compatible branch safe, yet no observation-adapted control can recover every branch once the distinguishing observation arrives; the finite-horizon recursion (Section 3.1) captures that mode, but no continuous-time certificate for it is supplied here. An explicit instance — a three-state system whose root belief is one-step viable yet

whose post-observation belief is nonviable, with no certificate firing — is given in the Supplementary Material (S3, A.3).

- (iii) The information-set semantics is set-membership; probabilistic or belief-space formulations require separate state-ments, although the common-action obstruction transfers mutatis mutandis to any semantics in which the policy must choose a single action per information state.
- (iv) The results are finite-dimensional and time-invariant; the delay-free setting makes the timing bound of Theorem 3 sharp, and delay systems require the retarded extension of the Dini comparison argument (Hale and Verduyn Lunel, 1993).
- (v) Theorem 4’s adversarial-realization step presupposes both the closed-graph regularity of Section 2.1 and the closed-loop existence hypothesis (H5.2) — D lower semicontinuous or constant, or the convexified reading; without these the certificate remains a heuristic.
- (vi) The institutional consequences of Section 6.4 are design conclusions from the certificates, not empirical findings; their empirical testing belongs to applied studies.

7. Complete Characterizations and the Residual Gap

The certificates of Section 3 are sound sufficient conditions for nonviability, and Section 6.5 recorded that they do not exhaust the complement of the epistemic kernel. This section makes that gap precise: it exhibits two settings in which the viability question admits an exact answer — the one-step (finite discrete) setting, where the common-action certificate is necessary and sufficient, and the static-observation setting, where the problem reduces exactly to open-loop robust viability — and states the residual dynamic gap as an open problem.

Theorem 5 (one-step completeness). Let X, A, D, Y be finite, with transition $x^+ = F(x, a, d)$, observation $y = O(x, a, x^+)$, and safe set $\mathcal{V} \subseteq X$. Define the one-step safe set

$$S_1(x) = \{a \in U(x) : F(x, a, d) \in \mathcal{V} \text{ for all } d \in D(x)\}.$$

A belief $B \subseteq \mathcal{V}$ is one-step viable under $\mathcal{I}$ — some observation-based policy achieves $x_1 \in \mathcal{V}$ for every $x_0 \in B$ and every admissible disturbance — if and only if $\bigcap_{x \in B} S_1(x) \neq \emptyset$. Hence $B \notin \mathcal{W}_1$ exactly when the common safe-action set is empty, the discrete reading of Theorem 2.

Proof. ($\Rightarrow$) An observation-based policy depends on the record only, and every $x \in B$ produced the same record, so the policy plays a single $a \in U^B(B)$; safety of the first step forces

$F(x, a, d) \in \mathcal{V}$ for all $x \in B$ and $d \in D(x)$, i.e. $a \in \bigcap_{x \in B} S_1(x)$. ($\Leftarrow$) Playing any a in the intersection keeps every successor in $\mathcal{V}$. $\square$

Combined with the backward recursion $\mathcal{W}_{k+1} = \text{Pre}(\mathcal{W}_k)$ of Theorem 1, Theorem 5 shows that the pair (*common-action certificate*, *recursion*) is a complete characterization at every finite horizon in finite systems: the certificate detects first-step emptiness, and the recursion propagates the obstruction to every later step.

Theorem 6 (static-observation reduction to open-loop viability). Let $\mathcal{V}$ be compact with C^1 constraint functions q_j , f continuous with D compact-valued and of closed graph, $U(x) \equiv U$, and suppose the observation is *static*: $y(t) = O(x(t))$ is constant along every admissible trajectory, so no information accrues over time. Then every observation-based policy coincides with an *open-loop* control — a measurable function of time only, $u(\cdot) : [0, \infty) \rightarrow U$ — and $B_0 \in \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ if and only if there exists an open-loop control under which every trajectory from every $x_0 \in B_0$ and every admissible disturbance realization stays in $\mathcal{V}$ for all time. Within the constant-control subclass $u(\cdot) \equiv u_0$, the condition is the robust Nagumo boundary condition

$$\nabla q_j(x) \cdot f(x, u_0, d) \geq 0 \quad \text{for all } x \in \partial\mathcal{V}, \text{ active } j, d \in D(x),$$

necessary for a constant control to keep $\mathcal{V}$ invariant and sufficient under the standard regularity assumptions (Aubin, 1991; Frankowska, 1989). A time-varying open-loop control may be viable when no constant control is; the open-loop reduction, not the constant-control condition, is the exact characterization in the static class.

Proof. A static observation makes the record constant, so a policy’s action at time t can depend on time only — open-loop; this gives the reduction in both directions (a viable policy yields a viable open-loop control, and every open-loop control is an admissible observation-based policy). The constant-control statement is the robust Nagumo theorem (Aubin, 1991; Frankowska, 1989): *necessary*, because any trajectory leaving $\mathcal{V}$ under u_0 exhibits a boundary point with outward drift, and *sufficient* under the regularity assumptions. $\square$

Theorem 6 makes the sense of completeness in the static class precise: the obstruction is exactly the nonexistence of a robustly viable open-loop control, and within the constant-control subclass the boundary condition of Theorem 2 is necessary and sufficient. Completeness fails only when information accrues, where the residual modes are post-observation recourse and timing — the hidden-mode example and the timing bound of Section 3.3.

Open Problem 1 (dynamic-observation completeness). Give a finite, checkable certificate that is necessary and sufficient for membership in $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$ under dynamic observation beyond the belief-space restatement below, or prove a separation showing that none exists in a natural class. The belief-space restatement is: $B_0 \in \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ if and only if B_0 belongs to the viability kernel of the set-valued belief dynamics under the observation constraint (Cardaliaguet, Quincampoix, and Saint-Pierre, 2007); this is a complete characterization on the infinite-dimensional space of compact beliefs, and the obstruction calculus of Section 3 is its finite, certifiable layer.

8. A Calibrated Case Study and a Coverage Audit

We instantiate the three observation-side mechanisms on a concrete resource model and audit the certificates against the exact recursion of Theorem 1. The model is a single-species stock

$S \in [0, 1]$ (fraction of carrying capacity) with discrete dynamics

$$\begin{aligned} S^+ &= \text{clip}(S + S(1 - S) - u + w), \\ u &\in [0, 0.3], \quad w \in \{-0.02, 0, +0.02\}, \end{aligned}$$

harvest u , recruitment noise w read adversarially, and a sustainability floor $S_{\min} = 0.3$ (Clark, 1990).

The fibre criterion. The regulator observes a coarse two-bin index, low if $S < 0.6$ and high if $S \geq 0.6$. The safe set $\mathcal{V} = [0.3, 1]$ is not a union of index bins: the low bin contains both 0.1 (below the floor) and 0.4 (above it). By Proposition 6 no exact observation-only certifier exists, and by Corollary 1 the certainly-safe set is the single bin $\mathcal{V}_{\text{safe}} = \{\text{high}\}$. A verdict of “safe” on the strength of a low reading is therefore unsound; the index can certify safety only when it reads high.

Two-dimensional aggregation. The fibre criterion also governs a multidimensional stock. Take a two-species state $(S_1, S_2) \in [0, 1]^2$ with the floor $S_1 \geq 0.4$, and suppose the regulator observes only the aggregate $I = S_1 + S_2$. The states $(0.5, 0.5)$ and $(0.2, 0.8)$ are both compatible with the reading $I = 1.0$, yet the first satisfies the floor and the second violates it, so by Proposition 6 no exact observation-only certifier exists; by Corollary 1 the certainly-safe readings are exactly $I \geq 1.4$ (the smallest S_1 compatible with I is $I - 1$, so $I - 1 \geq 0.4$). An aggregate reading below 1.4 must therefore fall silent, however carefully thresholded — the two-dimensional form of the aggregation consequence of Section 6.4.

The delayed-information timing bound. The timing obstruction is sharpest in the hidden-regime system of Section 3.3: regimes $\theta \in \{-1, +1\}$, dynamics $z^+ = z + \theta u$, controls $u \in \{-1, +1\}$, floor $z \geq 1$, with θ revealed only after a blind window of T_{obs} steps. From $z_0 = 2$ the drift of $q(z) = z - 1$ on the declining branch is -1 under every admissible action, so the threshold of Theorem 3 and Remark 1 is $\sigma^*(B_0) = z_0 - 1 = 1$: nonviability is certified for every $T_{\text{obs}} > 1$, while $T_{\text{obs}} = 1$ is viable — the regime is revealed in time, and $u = \theta$ then restores growth. In the fishery above the same certificate is silent in the normal regime, because $u_0 = 0$ keeps every stock above the floor; the bound fires only when no blind action avoids decline, which is precisely its scope.

The certainty-equivalence trap. With a biased index $\hat{S} = S + 0.1$, $g(S) = S^2$, and $\mathcal{V} = [1, 2]$, the certainty-equivalence policy $u = g(\hat{S})$ produces $\dot{S} = g(S + 0.1) - g(S) = 0.2S + 0.01 > 0$, which reaches the upper bound from $S = 1$ in ≈ 3.3 time units, while the bias-corrected policy $u = g(\hat{S} - 0.1)$ keeps $\dot{S} = 0$ and the kernel nonempty (Remark 3).

A reproducible delayed hidden-regime model. The audit runs on the exact instance specified here. The state is (z, θ) with $\theta \in \{-1, +1\}$ a hidden regime and $z \geq 0$; the action $u \in \{-1, +1\}$ is common to both regimes; the transition is $z^+ = z + \theta u$; the safe set is $\mathcal{V} = \{z \geq 1\}$; and the observation reveals nothing before T_{obs} and reveals θ exactly at time T_{obs} . From a known z_0 with both regimes possible, the belief after $k < T_{\text{obs}}$ blind steps contains the two extreme histories $(z_0 - k, -1)$ and $(z_0 + k, +1)$, and a blind policy must choose the same u for both branches at every blind step. The guaranteed blind-window survival time is therefore $\sigma^*(B_0) = z_0 - 1$: the best blind policy drives one branch down at unit rate, so the declining branch reaches the floor 1 at time $z_0 - 1$, and after the reveal $u = \theta$ keeps $z \geq 1$ forever. Hence B_0 is robustly epistemically viable if and only if $z_0 - 1 \geq T_{\text{obs}}$, i.e. $z_0 \geq 1 + T_{\text{obs}}$ — the ground truth against which the certificates are audited. The one-step certificate of Theorem 2 fires exactly when $z_0 < 2$ (one blind step already forces $z_0 - 1 < 1$), and the timing bound of Theorem 3 fires exactly when $z_0 \geq 2$ and $T_{\text{obs}} > z_0 - 1$.

A coverage audit. To quantify the gap identified in Section 6.5, we audit the certificates against the exact recursion on the delayed hidden-regime system, over the grid $z_0 \in \{1.0, 1.1, \dots, 2.5\}$, $T_{\text{obs}} \in \{1, 2, 3\}$. Table 2 and Figure 4 report the true verdict, the one-step certificate, and

the timing bound. The audit is reproduced by the script `paper2_coverage_audit.py` (archived with the verification code at <https://zenodo.org/records/22545740>), which also regenerates Table 2 and Figure 4.

Remark 4 (joint completeness in the delayed class). In every audited cell the true verdict is reproduced by exactly one of the two certificates: of the 42 nonviable cells, 30 are certified by the common-action certificate and the remaining 12 — precisely those with $z_0 \geq 2$ and $T_{\text{obs}} > z_0 - 1$ — by the timing bound; none is uncovered. In this class the two certificates are jointly complete, and the Section 6.5 gap reduces to the timing cells.

Why joint completeness holds in this class. The ground truth is the identity $B_0 \in \text{ERViab}_T(\mathcal{V}) \iff z_0 \geq 1 + T_{\text{obs}}$ established in the reproducible-model paragraph above, so nonviability holds exactly when $z_0 < 1 + T_{\text{obs}}$. The two certificates partition that set: the common-action certificate fires exactly when $z_0 < 2$, and the timing bound fires exactly when $z_0 \geq 2$ and $T_{\text{obs}} > z_0 - 1$, that is $2 \leq z_0 < 1 + T_{\text{obs}}$; the two conditions are disjoint and their union is $\{z_0 < 1 + T_{\text{obs}}\}$. Every nonviable cell is therefore certified by exactly one certificate, so within the delayed hidden-regime class the pair (common-action, timing) is a complete characterization at every horizon, and the gap of Section 6.5 is exactly the timing cells.

Calibration. The certificate parameters correspond to quantities estimable from standard stock-assessment practice: the drift margin ε is a net growth rate at the floor; the first informative observation time T_{obs} is the assessment or review periodicity; the bias b of the certainty-equivalence instance is the systematic error of the indicator; and the floor is the reference point of the management rule. Calibrating these four numbers from a published assessment is all the machinery the certificates require; the present section stays symbolic, and the empirical calibration is left to applied studies.

The audit is symbolic and one-dimensional (with the two-dimensional aggregation instance above); a multidimensional belief-space numerical campaign is deferred to future work, and the case study makes no claim of a general-purpose computational calculus.

9. A Probabilistic Development: Belief-State Safety Values

The certificates of Sections 3–4 rest on set-membership semantics: the information state is a set of compatible states. This section records the stochastic counterpart — the belief-state safety value of partially observable control — and its obstruction certificate, connecting the calculus to the partially observable literature (Åström, 1965; Smallwood and Sondik, 1973).

Let X, A, D, Y be finite, with a stochastic transition $T(x' | x, a)$ and an observation likelihood $g(y | x, a, x')$. Augment the state space with an absorbing unsafe state $\perp$: for $x \in \mathcal{V}$, $T(\perp | x, a) = 1 - \sum_{x' \in \mathcal{V}} T(x' | x, a)$, and $T(\perp | \perp, a) = 1$. A belief is a distribution b over $X \cup \{\perp\}$, updated by Bayes' rule $\tau(b, a, y)$, with $\mathbb{P}(y | b, a) = \sum_{x, x'} b(x) T(x' | x, a) g(y | x, a, x')$. The *safety value* is the maximal survival probability

$$V_k(b) = \max_{\pi} \mathbb{P}_{\pi}(x_t \in \mathcal{V} \text{ for } t = 0, \dots, k | b).$$

Theorem 7 (belief-state safety value). $V_0(b) = b(\mathcal{V})$ (the horizon counts the current state), and for $k \geq 0$

$$V_{k+1}(b) = \max_{a \in A} \sum_{y \in Y} \mathbb{P}(y | b, a) V_k(\tau(b, a, y)),$$

with the time convention that the policy acts, observes y , then acts again. The recursion is exact on every finite horizon; it is the Smallwood–Sondik belief-state value iteration specialized to the absorbing safety reward (Smallwood and Sondik, 1973).

Proof. Standard value iteration for the belief-state Markov decision process: conditioning on the observation after the action and applying the optimal continuation gives the identity, and the absorbing state makes V_k the k -step survival probability (Smallwood and Sondik, 1973). $\square$

Proposition 8 (chance-constrained common-action obstruction). Let $b \in \Delta(\mathcal{V})$ (so $b(\mathcal{V}) = 1$) and write $p(x, a) = 1 - \sum_{x' \in \mathcal{V}} T(x' | x, a)$. If

$$\sum_{x \in \mathcal{V}} b(x) p(x, a) \geq \delta \quad \text{for every } a \in A,$$

then $V_k(b) \leq 1 - \delta$ for every $k \geq 1$, and the chance constraint $\mathbb{P}_{b, \pi}(x_0, \dots, x_k \in \mathcal{V}) \geq 1 - \varepsilon$ is infeasible under every policy whenever $\varepsilon < \delta$.

Proof. The one-step survival probability under action a is $1 - \sum_x b(x) p(x, a) \leq 1 - \delta$, so $V_1(b) \leq 1 - \delta$. Since $V_{k+1} \leq V_k$ — more steps cannot raise the survival probability — $V_k(b) \leq V_1(b) \leq 1 - \delta$. $\square$

Proposition 9 (degenerate (deterministic) limit). Let X, A, D, Y be finite with the deterministic maps $x^+ = F(x, a, d)$ and $y = O(x, a, x^+)$ of Section 3.1, and let the stochastic transition and likelihood have supports $\{F(x, a, d) : d \in D(x)\}$ and $\{O(x, a, x^+) : x^+ \text{ reachable}\}$. Then the posterior support under the belief-state update is exactly the set-valued post-state $\text{Post}(B, a, y)$ of Section 3.1, so the support of the belief-state recursion follows the set-valued recursion $\mathcal{W}_k$. For a belief b carried by $B \subseteq \mathcal{V}$ and the disturbance read adversarially within its support, $V_k(b) = 1$ if $B \in \mathcal{W}_k$, while if $B \notin \mathcal{W}_k$ then $V_k(b) \leq 1 - \min_{x \in B} b(x) < 1$.

Proof. Under the deterministic kernels the posterior is supported exactly on $\text{Post}(B, a, y)$, so the support of the belief evolves by the set-valued recursion. If $B \in \mathcal{W}_k$, the witness actions of Theorem 1 keep every compatible branch in $\mathcal{V}$, giving $V_k(b) = 1$. If $B \notin \mathcal{W}_k$, every action leaves some compatible state $x \in B$ of mass $b(x)$ with a successor outside $\mathcal{V}$ under an admissible disturbance, so the one-step survival probability under every action is at most $1 - \min_{x \in B} b(x)$; since V_k is non-increasing in k , $V_k(b) \leq 1 - \min_{x \in B} b(x) < 1$. $\square$

Proposition 9 is the sense in which the chance-constrained obstruction degenerates to the common-action obstruction: at the deterministic limit $V_1(b) = 1$ exactly when some action keeps every branch of the support in $\mathcal{V}$, i.e. when the common safe-action set is nonempty. It is a finite-model statement; a continuous-state limit requires uniform concentration of the kernels, which is not developed here. The deficit from 1 is at least the smallest compatible mass $\min_{x \in B} b(x)$.

A worked instance (the two-floor conflict). Read the two-floor system of Section 3.4 as a finite POMDP: states x_1 and x_2 share one observation; the action $u \in [0, 1]$ is admissible; the one-step violation probability is $p(x_1, u) = 1$ for $u > 0.4$ and 0 otherwise, and $p(x_2, u) = 1$ for $u < 0.6$ and 0 otherwise; the prior is uniform, $b_0 = (1/2, 1/2)$. Then $\sum_x b_0(x) p(x, u) \geq 1/2$ for every u , so Proposition 8 gives $V_k(b_0) \leq 1/2$ for all $k \geq 1$, and the chance constraint $\mathbb{P}_{b, \pi} \geq 1 - \varepsilon$ is infeasible for every $\varepsilon < 1/2$. The bound is attained: $V_1(b_0) = 1/2$ (each action sacrifices exactly one floor), and the surviving floor is thereafter kept safe by its own action, so $V_k(b_0) = 1/2$ for all k . At the degenerate limit this reads $B_0 = \{x_1, x_2\} \notin \mathcal{W}_1$ with $V_k(b_0) = 1 - \min_{x \in B_0} b_0(x) = 1/2$, the bound of Proposition 9 — the chance-constrained obstruction evaluated at its exact value.

10. Conclusion

Under standard finite-dimensional regularity assumptions, viability under perfect measurement has a mature kernel, tan-

T_{obs}	initial position z_0																
	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	2.1	2.2	2.3	2.4	2.5	
1	●	●	●	●	●	●	●	●	●	●	○	○	○	○	○	○	
2	●	●	●	●	●	●	●	●	●	●	*	*	*	*	*	*	
3	●	●	●	●	●	●	●	●	●	●	*	*	*	*	*	*	

Table 2: Coverage audit on the delayed hidden-regime system: ○ viable; • nonviable, certified by the common-action certificate (Theorem 2); * nonviable, certified only by the timing bound (Theorem 3). Every nonviable cell is covered by exactly one certificate.

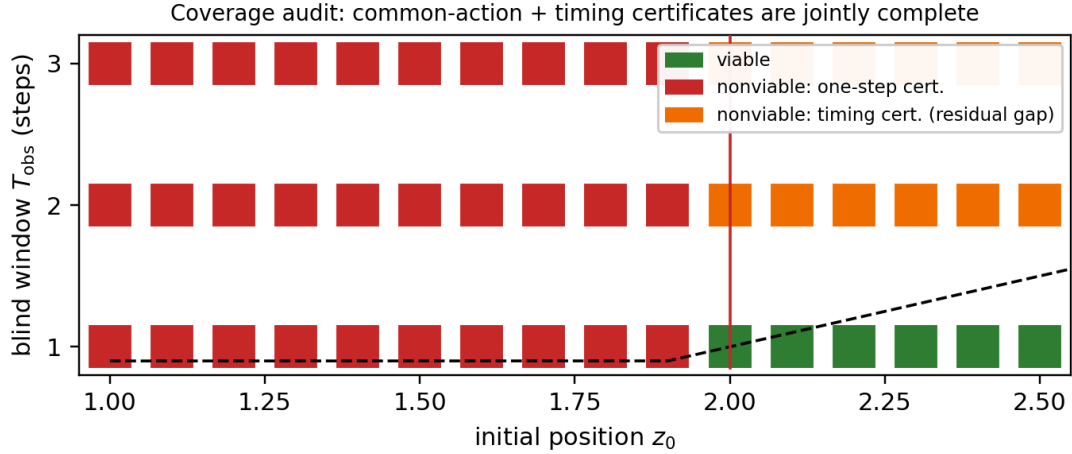


Figure 4: The coverage audit of Table 2: green = viable; red = nonviable, certified by the common-action certificate; orange = nonviable, certified only by the timing bound (the residual gap of Section 6.5). The dashed curve is the true boundary $z_0 = 1 + T_{\text{obs}}$ and the solid vertical line the one-step boundary $z_0 = 2$.

gency, and approximation theory. Viability under incomplete observation has long been asymmetric: the sufficiency side is carried by Veliov’s output-feedback condition and the estimation-tube programme, while the failure side has received less systematic treatment. This paper supplies the missing instrument: the obstruction calculus develops sound, computationally interpretable certificates for nonviability — that is, sufficient conditions for nonviability, equivalently necessary conditions for viability. They do not exhaust the complement of the epistemic kernel (Section 6.5), and the paper claims no complete characterization. The mechanisms are organized by one selector principle (Proposition 1) — observation-based viability is the existence of a common recursively safe response — and are nested in an obstruction ladder (Proposition 3); in finite systems backward belief recursion is sound and complete (Theorem 1), so the calculus is exact where it can be and sound elsewhere.

With displayed proofs and explicit witnesses, the certificates identify five mechanisms by which an observation structure can make robust viability impossible, plus one through which a single uncorrected policy empties the kernel: the disturbance can enforce exit in finite time against every control; a constant observation can merge states with incompatible admissible controls (Proposition 5’s admissibility form); the compatible states can demand incompatible actions at a single information state; the information can arrive after the enforced exit; and the observation fibres can cross the safe-set boundary, defeating every exact certifier — with the uncorrected certainty-equivalence controller as the sixth, policy-specific, mechanism. Each certificate that admits a design remedy identifies it: a faster indicator, a separating observation, an observation that separates the sides of each floor instead of aggregate certification, or the correction of a known bias. For sustainability governance, where observation is conducted through coarse indicators at discrete reviews, the calculus states the quantitative limits of

what monitoring can deliver — and converts each impossibility into a design specification. Three extensions complete the picture: the certificates are complete in the one-step and static-observation classes, with the residual dynamic gap stated as an open problem (Section 7); a calibrated case study and a coverage audit show the certificates reproducing the exact recursion verdict on a grid of delayed hidden-regime systems (Section 8); and the probabilistic belief-state lift connects the calculus to the partially observable control literature (Section 9).

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Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The coverage-audit script `paper2_coverage_audit.py` is archived at <https://zenodo.org/records/22545740>. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.